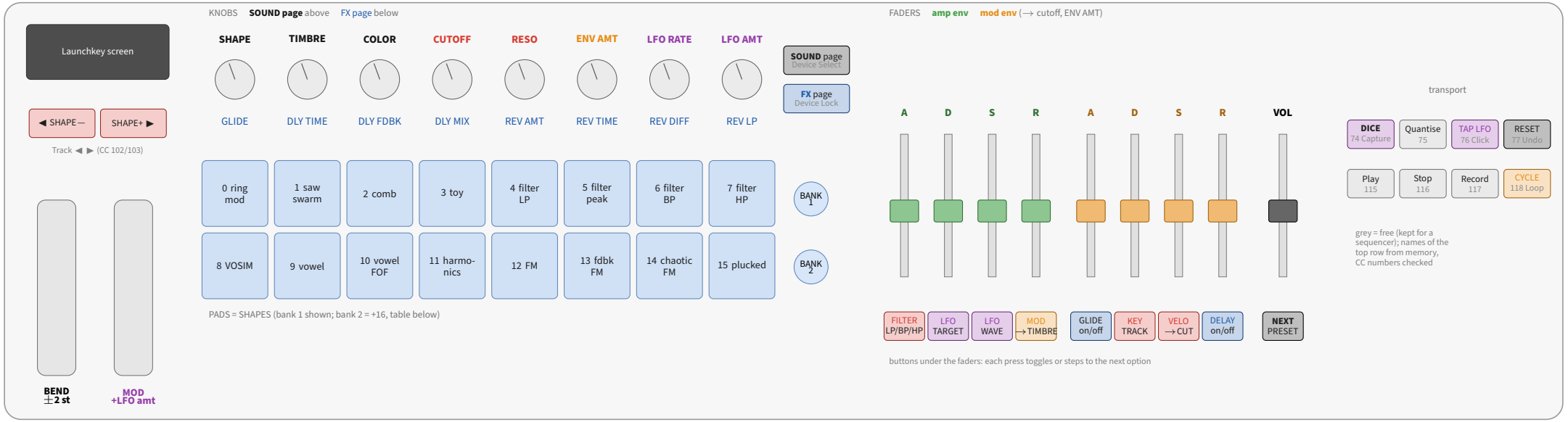


Knobs: two pages, scaled pickup

SHAPE	34 Braids shapes in turn (3–4 steps each)
TIMBRE / COLOR	the two Braids parameters (meaning depends on the shape)
CUTOFF	20 Hz $\cdot 2^{10k}$: 20 Hz – 20 kHz
RESO	Q = 0.5 – 20 (exponential)
ENV AMT	mod env → cutoff, ±5 octaves, 0 at centre
LFO RATE	0.05 – 20 Hz (or tap tempo)
LFO AMT	0 – 1, plus the mod wheel
GLIDE	0 – 1 s (quadratic), with GLIDE on
DLY TIME / FDBK / MIX	10 ms – 1.5 s; 0 – 0.95; 0 – 1
REV AMT/TIME/DIFF/LP	Elements reverb, 0 – 1 each

Page: Device Select = SOUND, Device Lock = FX (the two buttons right of the knobs). A press always lands on a known page. After a page change or RESET, a knob that is not where its value is still works: the value moves *proportionally* the way you turn and meets the knob at an end of travel at the latest. No jumps, no dead knobs. FX knobs 1–4 are silent until GLIDE or DELAY is on.

Envelopes and filter

Faders 1–4 **amp**, 5–8 **mod**: A, D, R = 1 ms $\cdot 10^{4k}$ (1 ms – 10 s, a decade per quarter of the travel); S = level 0 – 1. Exponential segments; attack aims at 1.2: a concave rise, fast at first (analog style).

Mod env → cutoff (ENV AMT), and → timbre (±0.5) with MOD→TIMBRE. **CYCLE** (Loop): it loops attack/decay while a key is held.

Filter: 12 dB/oct state-variable (LP/BP/HP, BP normalised). KEY TRACK (on at power-on): cutoff follows the keys, 1 oct/oct around note 60 (middle C). VELO→CUT: up to +2 octaves for a hard hit.

LFO (one, shared by the voices)

TARGET: **timbre** → color → cutoff (±3 oct) → pitch (±2 st). WAVE: **sine** → triangle → saw → square → S&H. TAP LFO: two taps 0.1–4 s apart set the period; LFO RATE takes over again when turned.

Pads: shapes, two banks

Bank 1 (0–15)
0 ring mod 1 saw swarm
2 comb 3 toy
4–7 filter LP/peak/BP/HP
8 VOSIM 9 vowel
10 vowel FOF 11 harmonics
12 FM 13 feedback FM
14 chaotic FM 15 plucked

Bank 2 (16–31)
16 bowed 17 blown
18 fluted 19 bell
20 drum 21 kick
22 cymbal 23 snare
24 wavetables 25 wave map
26 wave line 27 paraphonic
28 filtered noise 29 twin peaks
30 clocked noise 31 granular

Top row = first 8, bottom row = next 8. 32 particle noise and 33 digital modulation: ▶ from 31, or the SHAPE knob. Strike-type shapes (plucked, bell, drums) are struck on each key press.

Notes

3 voices (a 4th note takes the oldest voice). **NEXT PRESET** (fader button 9) steps through 8 presets (page 2); **RESET** (Undo) is preset 0, the power-on sound; **DICE** (Capture MIDI) rolls a random SOUND page and keeps envelope, FX and volume. Controls left behind by a preset or a roll pick up smoothly. Toggle states are **not shown** yet (no LEDs in standalone mode).

The Launchkey's own features

They make notes, so they play freak as well (set them on the Launchkey, see its user guide):

Arp: up, down, up/down, as played, random, chord, mutate; 1/4 – 1/32T; 1–4 oct; latch, gate, swing, rhythm.
Scale: minor, major, Dorian, Mixolydian, Phrygian, harmonic minor, minor/major pentatonic.
Fixed chord, octave ±, transpose.
Not yet tested with freak.

Signal path and load

3 × (Braids → SVF → amp env × velocity) → delay → Elements reverb → VOL. Velocity: soft notes keep $\frac{1}{4}$ of the level. DSP 57 %, patch RAM 40 of 44 KB (Axoloti Core, Ksoloti 1.1.0).

Standalone: no laptop

Save: in the patch window, *Upload to Internal Flash as Startup*, and also *Upload to SD Card as Startup*: at power-up the board looks on the SD card first, then in flash.
Play: power the Axoloti from a USB charger (micro-USB port); the Launchkey on the USB host port is powered by the board. freak starts by itself on preset 0; knobs and faders pick up where they are.
If it will not start: connect the laptop; `tools/axo_dfu -n` puts the board in Rescue Mode without opening the box, then *Board → Firmware → Flash (Rescue)*. Holding S2 at power-up skips the startup patch (S2 is inside the box).

The 34 shapes: what TIMBRE (knob 2) and COLOR (knob 3) do (after the Braids manual)

#	pad	shape	TIMBRE	COLOR
0	1·T1	**ring mod**	tuning of sines 2, 3	tuning of sines 2, 3
1	1·T2	**saw swarm**	detune (7 saws)	high-pass
2	1·T3	**comb**	comb pitch	feedback — / 0 / + (L / noon / R)
3	1·T4	**toy**	clock rate	glitches
4	1·T5	**filter LP**	cutoff	wave: saw→square→tri
5	1·T6	**filter peak**	cutoff	wave: saw→square→tri
6	1·T7	**filter BP**	cutoff	wave: saw→square→tri
7	1·T8	**filter HP**	cutoff	wave: saw→square→tri
8	1·B1	**VOSIM**	formant 1	formant 2
9	1·B2	**vowel**	vowel a e i o u	formant shift
10	1·B3	**vowel FOF**	vowel a e i o u	formant shift
11	1·B4	**harmonics**	centre of the 12 harmonics	their spread
12	1·B5	**FM**	mod amount	ratio
13	1·B6	**feedback FM**	mod amount (harsher)	ratio
14	1·B7	**chaotic FM**	mod amount (unstable)	ratio
15	1·B8	**plucked**	damping	pluck position
16	2·T1	**bowed**	friction	bow position
17	2·T2	**blown**	air pressure	geometry
18	2·T3	**fluted**	air pressure	geometry
19	2·T4	**bell**	damping	inharmonicity
20	2·T5	**drum**	damping	brightness
21	2·T6	**kick**	decay	tone
22	2·T7	**cymbal**	band-pass cutoff	drone ↔ noise
23	2·T8	**snare**	tone	snappy (noise)
24	2·B1	**wavetables**	sweep the table	table, 1 of 20
25	2·B2	**wave map**	X in a 16×16 map	Y
26	2·B3	**wave line**	position	smoothing (left: stepped)
27	2·B4	**paraphonic**	wave	**chord** (left: unison)
28	2·B5	**filtered noise**	resonance	LP ↔ HP; keys = cutoff
29	2·B6	**twin peaks**	resonance Q	peak spacing
30	2·B7	**clocked noise**	periodicity	quantise 2–32 steps
31	2·B8	**granular**	grain density	pitch randomness
32	▶	**particles**	density	pitch randomness
33	▶	**digital mod**	bit rate	data byte

pad = bank·row+column: 1·T1 is bank 1, top row, first pad (BANK 1/2 = the round buttons right of the pads). Plucked, bowed, bell, drum, kick, cymbal and snare are struck on each new key: S and R shape the tail only. Time constants inside Braids that do not follow pitch (grain rates, built-in decays) run at half the original speed.

Presets: NEXT PRESET (fader button 9) steps 0→7→0, RESET (Undo) = 0 (chosen from the parameter meanings, not yet tuned by ear)

0 init	saw swarm, LP, key tracking; delay off. The power-on sound.
1 vowel pad	vowel FOF; attack 0.5 s, release 1 s; LFO→timbre, sine, 0.2 Hz, 1/3 deep; cutoff 3.6 kHz; delay 0.12 s (fdbk 0.3, mix 1/4), reverb 1/2.
2 FM bass	FM, light modulation; plucky amp (decay 0.1 s, no sustain); low-pass at 200 Hz (note 60) opened +2.5 oct by the mod env; velocity→cutoff; 60 ms glide. Play it low.
3 glass mallets	bell, long ring, nearly harmonic; high-pass at 113 Hz; release 1 s; big reverb.
4 chord stabs	paraphonic: each key plays a chord (COLOR picks it); short stabs; delay 0.43 s, fdbk 0.48.
5 granular weather	granular, random grain pitch; band-pass; LFO S&H→color; CYCLE on: swelling, shifting texture.
6 pitched rain	particles, wide pitch spread; long delay feedback (0.71), big reverb.
7 chaotic drone	chaotic FM; attack 0.7 s, release 1.4 s; band-pass with a slow triangle LFO (0.13 Hz) on cutoff; CYCLE on; long delay and reverb. Hold a chord.

A preset changes both knob pages, faders 1–8, the shape and the toggles; the volume stays. **DICE** (Capture MIDI) rolls a random SOUND page instead. Controls that a preset or roll left behind pick up smoothly.

Where the values fall (knob or fader turned . . .)

	1/4	1/2	3/4	full
A, **D**, **R** (both envelopes)	10 ms	100 ms	1 s	10 s
CUTOFF (at note 60)	113 Hz	640 Hz	3.6 kHz	20 kHz
RESO (Q)	1.3	3.2	8	20
ENV AMT	−2.5 oct	0	+2.5 oct	+5 oct
LFO RATE	0.22 Hz	1 Hz	4.5 Hz	20 Hz
GLIDE	0.06 s	0.25 s	0.56 s	1 s
DLY TIME	35 ms	0.12 s	0.43 s	1.5 s
DLY FDBK	0.24	0.48	0.71	0.95
VOL (gain)	0.06	0.25	0.56	1

Times are exponential: each quarter of the travel is a factor of 10. S, TIMBRE, COLOR and the mixes are linear, 0–1.

Something seems wrong?

No sound: VOL (fader 9) up? CUTOFF (knob 4) not at the bottom in LP, nor at the top in HP? A struck shape (pluck, bell, drums) only sounds on a new key. Is the patch running (Live / startup)?

Knobs do nothing: on the FX page? Press Device Select. FX knobs 1–4 need GLIDE / DELAY on.

Wobbles or detunes: LFO AMT (knob 8) or the mod wheel; TARGET may be on pitch.

A note drops out: 3 voices; a 4th note takes the oldest.

Echoes pile up: lower DLY FDBK or switch DELAY off (the tail dies away).

Lost: RESET (Undo) restores the power-on sound.

With the patcher connected, its log prints page changes, resets, knobs catching up and unknown messages.

MIDI reference (Launchkey 49 MK3, standalone)

keys, bend, mod wheel	ch 1: notes 36–84, pitch bend, CC 1
knobs 1–8	ch 1 CC 21–28
faders 1–9	ch 1 CC 71–79
fader buttons 1–9	ch 1 CC 11–19 (127 press, 0 release)
pads	ch 10 notes 36–51
Device Select / Device Lock (pages)	ch 16 CC 51/52 (observed)
track ◀/▶	ch 16 CC 102/103
buttons left of the pads	ch 16 CC 106/107, also switch pages
round buttons, Capture, Click, Undo, Loop	ch 16 CC 104/105, 74, 76, 77, 118

Recorded from this unit (`docs/launchkey49mk3_map.md`).

Sound design moves

Pluck: mod env A 0, D 1/2 (100 ms), S 0; ENV AMT 3/4; CUTOFF 1/4. Every note snaps the filter open.

Second LFO: CYCLE (Loop) turns the mod envelope into a looping AD; its A and D faders set the rate.

Stepped motion: LFO WAVE to S&H, TARGET to timbre or color, then tap the tempo with Click.

Bigger from one key: shape 27 plays a chord per voice (COLOR picks it); the Launchkey's Fixed Chord does the same for any shape.

Explore: DICE until something sticks, shape it with the knobs, RESET if lost. Scale mode on the Launchkey keeps random sounds in key.